

Lionel Sopgoui

POSTDOCTORAL RESEARCHER IN MATHEMATICAL FINANCE : ENSAE PARIS

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Summary

I am an ingenious, curious, and dynamic young person with a passion for Quantitative Finance and Trading. I am constantly looking for new challenges, motivated by self-satisfaction to bring concrete and efficient solutions. I recently completed my PhD in Mathematical Finance. I am currently a postdoctoral researcher at ENSAE Paris. I work *Prof. Caroline Hillairet* and *Prof. Olivier Lopez* at ENSAE Paris. We use different mathematic tools such as Hawkes processes, contagion and epidemiological models, extreme value theory, etc. to model and quantify cyber risks for cyber insurance..

Education and Training

Université de Paris - Imperial College London, PhD student in Financial Mathematics	Sep. 2021 - Nov. 2024	Paris, France
Ecole Polytechnique - ENSAE Paris, M.S. in Financial Mathematics : Statistics and Finance	Sep. 2019 - Dec. 2020	Palaiseau, France
Telecom SudParis, Engineering Degree in Statistical Modelling and Applications	Sep. 2017 - Dec. 2020	Evry, France
Lycée Jeanne d'Albret, Preparatory Class for High Scientific School	Sep. 2016 - Jul. 2017	St Germain, France
Lycée Jacques Amyot, Preparatory Class for High Scientific School	Sep. 2015 - Jul. 2016	Melun, France
National Advanced School of Engineering, Mathematics and Computer Science	Sept. 2012 - July. 2015	Ydé, Cameroon

Work Experience

Bonetti Arbitrage Lab

CEO AND QUANTITATIVE TRADER

Paris, France

Nov. 2021 - Ongoing

- I am developing trading strategies based on statistical arbitrage on US small and mid-caps, and executing them on Interactive Brokers.
- My strategies are intraday, semi-systematic (Python and IBKR API), and give a triple-digit performance yearly as well as sharpe ratio greater 2.

Groupe BPCE

QUANTITATIVE RISK RESEARCHER

Paris, France

Mar. 2021 - Aug. 2024

- I am conducting research on the projection of portfolio risk measures for very long time horizons with application to climate transition risk.
- I am also incorporating the greenhouse gases costs into economic modelling, firm valuation, real estate, as well as credit risk modelling.

LexiFi

QUANTITATIVE RESEARCHER INTERN

Paris, France

May. 2020 - Nov. 2020

- I have used Deep Learning to predict and rebuild Implied Volatility Surfaces from historical spot and forwards series, for illiquid, proprietary or OTC options.
- I developped a new GAN called CRS GAN to generate Implied Volatility Sufaces, essential for Options Pricing and Portfolio scenario analysis.

Planète-Urgence

VOLONTEER

Eboudjé, Cameroon

Aug. 2019 - Sep. 2019

- I learned about Tubè Awù, an Association which takes care of marine biodiversity, especially marine turtles in Central Africa Coast.
- I helped Tubè Awù to improve its visibility and its communication by creating a Website, a Facebook page and a YouTube channel.

BlackRock

SUMMER INTERN

Budapest, Hungary

Jun. 2019 - Aug. 2019

- I read scientific publications on Natural Language Processing, then on Machine Learning Classification algorithms.
- I built, then proposed to my team a strong financial documents classifier, with an accuracy of 80%, based on a training set of over 4000 features.

Projects

I am implementing Trading Algorithms on Stocks (via IBKR API) and on Cryptos (via Binance API)

PERSONAL PROJECT

Paris, France

Jan. 2020 - Ongoing

Impact of the carbon price on credit portfolio's loss with stochastic collateral

THIRD PHD PROJECT | UNIVERSITÉ PARIS CITÉ - IMPERIAL COLLEGE LONDON

Paris, France

June. 2023 - Aug. 2024

Bouveret, Géraldine and Chassagneux, Jean-françois and Ibbou, Smail and Jacquier, Antoine and Sopgoui, Lionel, Propagation of a Carbon Price in Credit Portfolio Through Macroeconomic Factors (July 25, 2023). Available at SSRN:4518564

Modeling the impact of Climate transition on real estate

SECOND PHD PROJECT | UNIVERSITÉ PARIS CITÉ - IMPERIAL COLLEGE LONDON

Paris, France

Jan. 2024 - Aug. 2024

Sopgoui, Lionel, Modeling the impact of Climate transition on real estate (August 01, 2024). Available at SSRN:4913777

Propagation of a Carbon Price in Credit Portfolio Through Macroeconomic Factors

FIRST PHD PROJECT | UNIVERSITÉ PARIS CITÉ - IMPERIAL COLLEGE LONDON

Paris, France

Sept. 2021- July. 2023

Sopgoui, Lionel, Impact of the carbon price on credit portfolio's loss with stochastic collateral (August 12, 2024). Available at SSRN:4922966

Skills

Key strengths	Team player, Proactive problem solver, Detail oriented, Curiosity, Efficient under pressure, Highly self-disciplined.
Programming skills	Python (advanced), VB.NET (advanced), C++(advanced), J2EE (advanced), Java, Matlab, MySQL, R , SAS
Software & Tools	Tensorflow 2, Keras, LexiFi Apropos, Python Notebook, Visual Studio, Eclipse, Matlab, R Studio, Windows, Linux
Languages	French (Native), English (Advanced), Russian (Basic)

Interests and Activities

Writer	Les Essais de Paukémil (a political philosophy book published in January 2021), and a few articles in Mathematics and Philosophy.
Curious reader	Economy (On the Principles of Political Economy and Taxation), Sociology (Capital & Ideology), History, Philosophy, etc
Regular sportsman	Regular runner, Basketball, Tennis player and supporter.
World Citizen	Travelled to Cameroon, France, Germany, Hungary, Switzerland, Luxembourg. Discovered the world through gastronomy.